

AMENDMENT NO. 2 JANUARY 2009
TO
IS 13010 : 2002 ac WATTHOUR METERS,
CLASS 0.5, 1 AND 2 — SPECIFICATION

(First Revision)

(Page 4, clause 3.43) — Delete.

(Page 4, clause 5.2, line 3) — Substitute ‘for example’ for ‘that is’.

[Page 7, clause 6.6(a)] — Delete and renumber item ‘b)’ as ‘a)’.

(Page 8, clause 6.11, Title) — Substitute the following for the existing:

‘Direction of Rotation, Marking of the Rotor and Reverse Running Stop’

(Page 8, clause 6.11) — Insert the following new para after para 2:

‘When reverse running stop is provided to prevent rotation of the rotor contrary to the specified direction, the device shall be capable of restricting such rotation to less than one revolution when maximum current at unity power factor is applied 180° out of phase in each of the metering elements simultaneously. Moreover, as soon as such currents cease to exist, the rotor shall fulfil the requirements of starting in the specified direction.’

(Page 8, clause 7) — Substitute ‘**TORQUE/SPEED RATIO**’ for the existing title.

(Page 8, clauses 7.1 and 7.2) — Substitute the following for the existing clauses:

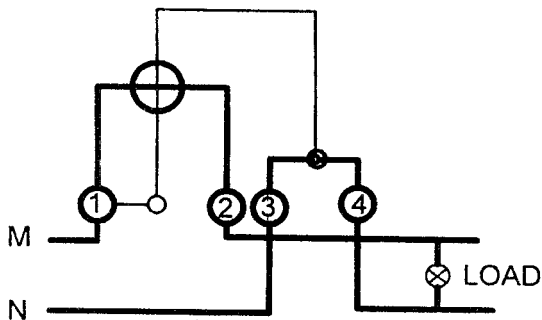
‘7.1 The nominal torque (T_b) of a meter at reference voltage, unity power factor and basic current expressed in gm-cm (*see 11.22*), and its speed (S_m), determined from meter constant, at reference voltage, unity power factor and maximum current, expressed in rpm, shall satisfy the following relationship:

$$\begin{aligned} T_b \times 100/S_m &\geq 5.0 \text{ for single-phase 2-wire} \\ &\geq 13.75 \text{ for three-phase 4-wire} \end{aligned}$$

NOTE — 1 gm-cm = 9.80665×10^{-5} newton-meter.

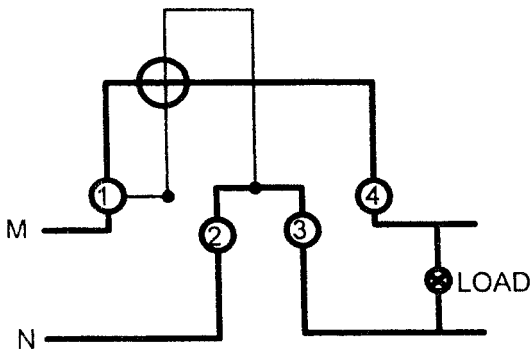
(Page 9, clause 8.1, Note 4, line 1) — Substitute ‘’ for ‘’.

[Page 9, clause 8.2.3, Fig. 1(a)] — Substitute the following diagram for the existing diagram:



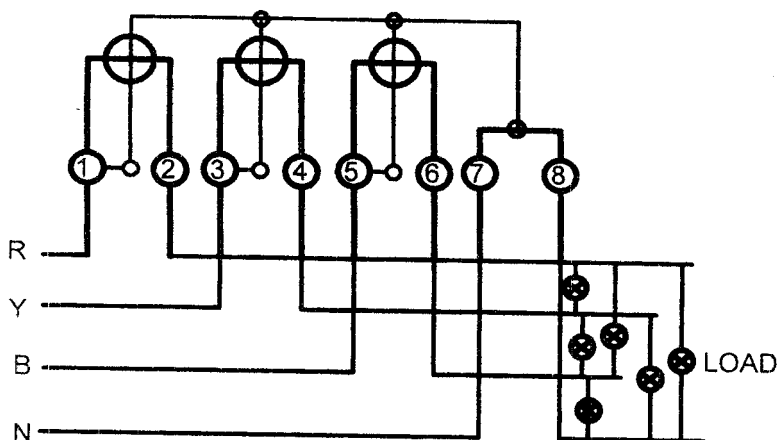
1a) Single-Phase Two-Wire Meter

[Page 9, clause 8.2.3, Fig. 1(b)] — Substitute the following for the existing diagram:



1b) Single-Phase Two-Wire Meter Symmetrical Connections

[Page 10, clause 8.2.3, Fig. 1(f)] — Substitute the following for the existing diagram:



1f) Three-Phase Four-Wire Single Tariff Meter for Direct Connection

(Page 11, Table 8, col 3, 4 and 5) — Substitute '+/-' for '+' wherever appears.

(Page 12, Table 9) — Add the following row after SI No. 2 and renumber the subsequent SI No.:

SI No. (1)	Test (2)	Clause Reference (3)
3	Requirement of prevention of reverse rotation (if provided)	6.11

(Page 12, Table 9) — Add the following row after renumbered SI No. 22 and renumber the subsequent SI No.:

SI No. (1)	Test (2)	Clause Reference (3)
23	Torque/Speed Ratio	11.22

(Page 12, Table 9) — Add the following row after renumbered SI No. 25:

SI No. (1)	Test (2)	Clause Reference (3)
26	Verification of general requirements and markings	6 and 8

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(Page 13, Table 10) — Add the following row after SI No. 2 and renumber the subsequent SI No.:

SI No. (1)	Test (2)	Clause Reference (3)
3	Requirement of prevention of reverse rotation (if provided)	6.11

(Page 13, Table 10) — Add the following row after renumbered SI No. 10:

SI No. (1)	Test (2)	Clause Reference (3)
11	Verification of general requirements and markings	6 and 8

(Page 18, clause 11.18, para 2, line 6) — Substitute ‘10 percent lb UPF,’ for ‘5 percent lb UPF,’.

(Page 19, Table 25, col 3) — Substitute ‘1’ for ‘2’.

[Page 19, clause 11.20(d)] — Substitute the following for the existing:

‘d) Acceleration (between 16.7 Hz and 150 Hz) : 2.2 g (g = acceleration due to gravity)’.

[Page 19, clause 11.20(f)] — Substitute the following for the existing:

‘f) Duration : One hour in each direction.

‘g) Direction : Along three mutually perpendicular axes (in the directions of upper and lower, right and left, back and forth)’.

(Page 20, clause 11.22, line 6) — Substitute ‘torque/speed ratio as specified in clause 7’ for ‘efficiency/permanence factor specified in Annex D’.

(Page 23, Annex D, Items 18 and 19) — Delete.

(ET 13)

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